

Ben Bubnick

HEALTHCARE DATA STRATEGY · DATA GOVERNANCE · ANALYTICS DELIVERY

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Summary

Healthcare Data Consultant with 10+ years of experience leading the technical delivery and governance of analytics, data platforms, and ML solutions across clinical, operational, and revenue-cycle domains. Combines hands-on expertise in Python, SQL, Snowflake, SQL Server, AWS, Azure, and production data pipelines with leadership in data quality, source-to-target reconciliation, documentation, and shared standards. Skilled at turning ambiguous healthcare challenges into trusted analysis, reusable data models, real-time services, and stakeholder-ready reporting supported by testing, CI/CD, and observability. Known for aligning engineering, data science, operations, and business stakeholders while establishing transparent, scalable, and well-governed solutions for regulated environments.

Experience

HHaEXchange

Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase/GitHub Actions, FHIR/Aidbox

HEALTHCARE DATA INTEGRATION CONSULTANT — CONTRACT VIA VEERSA TECHNOLOGIES

Oct. 2025 - Jun. 2026

Led healthcare source analysis, data-quality investigation and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered Snowflake models with dbt and SQL/Jinja to standardize operational healthcare data from [N source tables] across [N organizations], reducing client onboarding time by [X%].
- Implemented reusable SQL/Jinja logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains, reducing client-specific code by [X%].
- Built automated dbt and Snowflake QA suites for duplicates, nulls, referential integrity, and source-to-target reconciliation; traced record-level failures to root causes and reduced resolution time by [X%].
- Profiled source schemas and converted ambiguous business rules into documented mappings, acceptance criteria, and onboarding standards, enabling [N integrations] with [X%] fewer clarification cycles.

Adonis

AWS: S3, Snowflake, SQLMesh, SQL, Python

HEALTHCARE REVENUE CYCLE DATA CONSULTANT — CONTRACT VIA HEALTH DATA MOVERS

Mar. 2025 - Jul. 2025

Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding and reporting.

- Architected Snowflake and SQLMesh models for payer and revenue-cycle data, mapping [N source entities] across [N clients] to a reusable analytics schema and reducing onboarding time by [X%].
- Developed reusable SQL transformations and automated release gates for row counts, nulls, referential integrity, and reconciliation, achieving [X%] source-to-target agreement across [N records].
- Implemented schema-drift detection, audit logging, CI checks, and deterministic recovery procedures, reducing data regressions by [X%] and recovery time from [A hours] to [B hours].
- Authored source-to-target mappings and integration runbooks that standardized technical reviews, reduced mapping approval time by [X%], and supported [N client implementations].

GrowData Analytics

Ruby, RubyLLM, Tesseract OCR, GPT 4o mini, Cladue Opus/Sonnet, Rubocop, RSpec

INDEPENDENT DATA AUTOMATION & ANALYTICS CONSULTANT

Sep. 2024 - Oct. 2025

Built a document-to-data product that converted semi-structured invoices into validated, auditable transaction records.

- Engineered a Ruby, Tesseract OCR, and LLM pipeline that converted handwritten and printed invoices into structured transactions, using cross-field validation, reconciliation, and confidence routing to achieve [X%] extraction accuracy across [N invoices].
- Added RSpec-backed validation, structured audit logs, error taxonomies, retry controls, and usage reporting, reducing unrecoverable processing failures by [X%] and mean time to resolution by [Y%].

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

SENIOR DATA ANALYST

Jun. 2023 - Jan. 2025

Delivered healthcare analytics, recurring reporting and data-quality investigations while improving the governed data foundation used by the BI team.

- Used SQL across Athena and Redshift to reconcile membership and demographic reporting, isolating differences in extraction criteria, eligibility rules, distinct-member logic, and duplicate handling across [N members] and reducing reporting discrepancies by [X%].
- Established a governed repository for scheduled reports, ad hoc SQL, dbt tests, and Jupyter notebooks; introduced peer-review and documentation standards that reduced duplicated logic by [X%] and analyst onboarding time by [Y%].

Merative

Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps

SENIOR DATA SCIENTIST

Jul. 2022 - Mar. 2023

Led data validation, compliance and migration assurance for a regulated healthcare platform modernization.

- Developed automated Python and SQL regression tests for Azure Data Factory and Synapse pipelines, detecting schema drift, transformation defects, and record-level fidelity issues before release and reducing migration defects by [X%].
- Designed and executed HIPAA-compliant de-identification validation across more than 200 million patient records in Azure, verifying [N PHI fields and rules] and resolving [X%] of exceptions before HITRUST readiness review.

IBM

Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle

SENIOR DATA SCIENTIST

Nov. 2021 - Jul. 2022

Delivered client analytics and large-scale migration validation for healthcare administrative data.

- Built Spark, Hive, and Impala analytics for administrative healthcare data, translating [N business requirements] into validated reports and analytical outputs while reducing delivery time by [X%].
- Developed distributed migration-validation analyses across more than 150 billion patient records, identifying [N data-fidelity defects] before production transition and preventing [X downstream incidents].

SENIOR DATA ARCHITECT

Dec. 2020 - Nov. 2021

Led clinical-data integration and unstructured-text analysis at enterprise scale.

- Architected Spark and Hadoop pipelines that integrated and normalized more than 100 billion structured and semi-structured clinical records for [N downstream analytics and application workflows], meeting [X%] of processing SLAs.
- Built Python and Spark NLP workflows to analyze nearly six million provider notes, extracting [N reusable themes or features] and reducing manual text-review effort by [X hours or X%].

DATA ARCHITECT

Nov. 2017 - Dec. 2020

Owned client-facing integration, discrepancy resolution and data-quality improvement across multi-organization healthcare networks.

- Designed standardized SQL validation frameworks and client-review gates that reduced validation rework by 25% and mapping rework by 45% across [N implementations].
- Traced data and application discrepancies across a 73-organization healthcare network from source ingestion through production, documenting defects in Jira and reducing mean time to resolution by [X%].

SOFTWARE DEVELOPER, DATA PLATFORM

Sep. 2016 - Nov. 2017

Built internal data-curation and knowledge-transfer tools that reduced costs and onboarding effort.

- Built an internal data-curation application that replaced a \$30,000 annual vendor contract, then rewrote technical runbooks to reduce knowledge-transfer time by 80% for [N engineers or client teams].

DATA SCIENTIST

Nov. 2015 - Sep. 2016

Coordinated client integration delivery, contractor work and process automation.

- Directed technical delivery across 36 client data-integration projects and established code-review, QA, and issue-resolution standards for 12 remote contractors, improving on-time acceptance by [X%] and reducing escaped defects by [Y%].
- Developed Python, Ruby, and SQL automation that eliminated 28% of unnecessary development work; authored more than 125 technical knowledge-base articles, approximately 30% of the team's online content, to standardize implementation practices.

Explorys

Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENCE CONSULTANT — CONTRACT VIA TEKSYSTEMS

Jun. 2015 - Nov. 2015

Supported clinical-data integration and population analytics across more than 10 client data warehouses.

- Diagnosed production Hadoop and Oracle ingestion failures and implemented Jenkins-backed scheduling controls across more than 10 client data warehouses, increasing successful job completion to [X%] and reducing recurring incidents by [Y%].

AmTrust Financial Services

VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL

SOFTWARE ENGINEER

Mar. 2014 - Jun. 2015

Developed MVC components and web forms for insurance policy software.

- Designed and implemented C and ASP.NET MVC components and web forms for an insurance policy platform and new product offering, delivering [N production features] with [X%] fewer post-release defects.
- Profiled more than 65 rating forms, isolated C and SQL performance bottlenecks, and reduced median processing time from [A seconds] to [B seconds].
- Established automated unit testing for multi-tier C and ASP.NET MVC components, increasing code coverage from [A%] to [B%] and reducing regression defects by [X%].

PPG Industries

VB.NET, C#, SQL, C++, FORTRAN

COLOR SCIENTIST

Apr. 2012 - Jun. 2013

Developed numerical methods and software for automotive color matching and measurement.

- Developed C++ and FORTRAN numerical classification algorithms that improved automotive color-matching accuracy by 40%, validated against [N laboratory samples].
- Engineered VB.NET, C, and SQL software to correlate measurements across [N instruments], reducing inter-instrument variance by [X%] and improving QA repeatability by [Y%].
- Developed numerical formulation and pigment-analysis methods for an automated automotive color-matching facility, processing [N formulations per day] and reducing manual analysis by [X hours per week].

Multiple Institutions

ADJUNCT PROFESSOR

Aug. 2010 - May. 2019

Taught physics & astronomy lecture series and lab courses.

- Designed and delivered quantitative physics and astronomy curricula for [N courses and N students], using interactive, peer-led, and just-in-time methods to improve assessment scores by [X%].
- Produced more than 700 hours of recorded lectures and laboratory material using [LMS and recording tools], enabling on-demand instruction for [N students] with a [X%] completion rate.
- Analyzed evaluation and learning-outcome data to refine course delivery, exceeding institutional benchmarks by 16%-17% year over year.

Professional Recognition

HDM, SPECIAL RECOGNITION (2025), Recognized for exceptional performance and commitment in client delivery.

Arkos, ARKOS HEALTH BI STAR (2023), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.

Merative, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (2022), Recognized for systems migration efforts.

IBM, GOLD CHAMPION (2018 - 2022), Recognized for 1,477 classroom hours delivered and 43 certificates earned.

IBM, LAB SERVICES AWARD (2018), Recognized for coordinating internal teams to deliver solutions on time.
IBM, EMINENCE AND EXCELLENCE AWARD (2017), Recognized for developing a new data curation tool.
PPG, COLOR LAB AUTOMATION AWARD (2012), Recognized for expanding data collection and analysis reporting.

Volunteer

HSEA

ETL BRIGADE

Feb. 2025 - May. 2025

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

CSforAll

INSTRUCTOR

Oct. 2018 - May. 2019

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE

COACH

Mar. 2017 - Mar. 2019

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America

ETL GUILD

Jan. 2015 - Jun. 2015

The initiative was brought about to apply data science tools to improve and expand public record access.

Talks

AI Hackathon 2020: Surgery Scheduling Optimization

Dec. 2020

"An algorithm uses past utilization and current waitlist info to predict surgery duration.

IBM

REACH Initiative: IBM Watson Care Insights

2019

Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow.

IBM

Content & Data Analytics Learning Exchange

2018

Introduction of implementation of machine learning algorithms using real-world health data analytics.

IBM

2018 Innovation Research Interchange Fall Networks Conference

Sep. 2018

Using NLP to extract insight from unstructured data and transform it from a data lake.

IRI

Cross Team Training Series

2016

A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes.

EXPLORYS

Publications

Reproducible Postgres load of Tuva seed datasets

Aug. 2025

Reproducible Postgres load of Tuva seed datasets with robust validation.

HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction ^a

Dec. 2020

Improvement on traditional fixed ratio methods for total procedure time estimation.

IBM

Process and Requirements for CCD Data Integration

Oct. 2017

Processes and requirements for overcoming the many difficulties with CCD data curation.

IBM

AI Paint Formula Prediction from Spectroscopic Measurements ^b

May. 2015

Naive bayesian color and effects matching algorithm of complex automotive paint formula.

SHINYAPPS.IO

Characterization of Surface Coatings using RT Models for Color Matching

Feb. 2013

Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching.

USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy ^c

Dec. 2010

Spectroscopic and Photometric analysis of two open clusters in the galactic plane.

OHIOLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107	May. 2008
<i>Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX.</i>	VIZIER
Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107	Feb. 2008
<i>New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107).</i>	PASP
A Novel Method for Low-Toxic Photo-Etch Printing^d	May. 2003
<i>Application of aquatint etching techniques to photo-etch procedures.</i>	MAPC

Education		
2022	IBM Data Science Professional Certificate^a	IBM
2015	Certificate of Specialization Data Science ^b	Johns Hopkins University
2010	Master of Science Astrophysics ^c	University of Cincinnati
2004	Bachelor of Fine Arts Printmaking ^d	Ohio University